

Smart City Traffic Intelligence

Part 3: Machine Learning and AI-Powered Mobility

Final Capstone Report

1. Executive Summary

This project develops a machine-learning-based traffic intelligence solution using historical traffic observations from the westbound I-94 corridor near Minneapolis–St. Paul, Minnesota.

The objective of Part 3 is to extend the data analytics and Python engineering work completed in Parts 1 and 2 into an integrated AI solution for traffic analysis, prediction, travel-time recommendations and simulated deployment.

The project covers supervised machine learning, unsupervised learning, deep learning and explainability, advanced AI techniques, MLflow experiment tracking, a traffic recommendation system, MLOps deployment simulation, and responsible AI considerations.

The Random Forest Regressor achieved a Mean Absolute Error (MAE) of **271.3004** and an R^2 of **0.9442**, outperforming the Linear Regression baseline. The selected model was subsequently reconstructed and deployed through a simulated FastAPI prediction endpoint, with MLflow supporting experiment tracking and model governance.

The recommendation system uses historical traffic patterns, day type and weather conditions to identify suitable lower-traffic travel periods. It also includes safeguards for insufficient weather observations and generates plain-language recommendations.

Finally, the project evaluates data limitations, proxy-label risks, model fairness, human oversight and computational sustainability. No real accident dataset was used; accident risk was represented through a documented proxy based on traffic-volume-derived congestion and weather conditions.

Overall, Part 3 demonstrates how historical traffic data can support an integrated, explainable and operationally oriented AI decision-support prototype, while recognizing the limitations that must be addressed before real-world deployment.

2. Dataset and Preparation

2.1 Dataset Description

The project uses the **Metro Interstate Traffic Volume** dataset, containing historical hourly traffic observations from the westbound I-94 corridor near Minneapolis–St. Paul, Minnesota.

The processed dataset contains **48,187 observations and 30 columns**, covering the period from **October 2, 2012, to September 30, 2018**.

The dataset includes traffic volume, weather conditions, temperature, precipitation, cloud coverage, holidays and timestamps.

Because the data represent a single traffic corridor, the results are specific to this location and should not be generalized to an entire road network.

2.2 Data Preparation and Feature Engineering

The data preparation and feature engineering processes developed in Part 2 were reused in Part 3 to ensure consistency and reproducibility.

The engineered features include:

- Temporal features such as hour, day of week and weekend indicators.
- Cyclical time representations using sine and cosine transformations.
- Weather indicators for rain and snow.
- One-hot encoded weather categories.
- Normalized temperature and cloud-coverage features.
- Congestion categories derived from observed traffic volume.

For the Random Forest regression deployment model, **25 numerical predictor features** were used. The target variable was `traffic_volume`.

The categorical holiday field was converted into a numerical `is_holiday` indicator. Non-predictive fields and the target-derived `congestion_category` were excluded from the regression inputs to avoid inappropriate feature use and target leakage.

2.3 Model Evaluation Preparation

The regression data were divided into training and testing subsets using an **80/20 random split**, with `random_state=42`.

This produced:

- **38,549 training observations**
- **9,638 testing observations**

The same data structure and model configuration were used to reconstruct the Random Forest deployment candidate in Task 6.

Although random splitting provides a reproducible basis for comparing models, it does not fully represent future deployment conditions. Time-based validation is recommended before operational use.

2.4 Data Limitations

The dataset has important geographical and temporal limitations.

It represents one westbound traffic corridor and covers historical observations from 2012 to 2018. It does not include live traffic information or actual accident outcomes.

Consequently, the models should be interpreted as historical traffic intelligence tools rather than real-time, city-wide traffic or accident prediction systems.

3. Supervised Machine Learning

3.1 Objective

Task 1 applied supervised machine learning to two traffic intelligence problems:

- **Classification:** Classifying traffic conditions using a documented accident-risk proxy.
- **Regression:** Predicting hourly traffic volume using historical traffic and engineered features.

The models used the feature engineering developed in Part 2, including temporal, weather and holiday-related information.

3.2 Classification Using a Proxy Accident-Risk Label

Because actual accident records were unavailable, a proxy classification target was constructed using congestion and weather conditions.

Congestion levels were derived from quartiles of observed traffic volume. The high-risk proxy identified conditions where high or severe congestion coincided with severe or low-visibility weather.

The classification task was designed to demonstrate supervised classification and model evaluation, rather than to estimate verified accident probabilities.

Classification performance should be assessed using accuracy, precision, recall, F1-score and ROC AUC. In particular, precision and recall are important because the high-risk proxy may be less frequent than the other class.

The proxy label has an important methodological limitation: it is constructed from predefined congestion and weather conditions. Consequently, high classification performance would demonstrate the ability to reproduce this proxy definition, not necessarily the ability to predict real-world accidents.

Classification Model Evaluation Results

Two classification algorithms were trained and evaluated using the documented proxy accident-risk label.

Evaluation Metric	Logistic Regression	Random Forest Classifier
Accuracy	0.9656	0.9851
Precision	0.7747	0.9076
Recall	0.9798	0.9660
F1-score	0.8653	0.9359
ROC AUC	0.9937	0.9966

The **Random Forest Classifier** achieved the strongest overall classification performance, with an accuracy of **98.51%**, an F1-score of **0.9359**, and a ROC AUC of **0.9966**.

Logistic Regression achieved slightly higher recall (**0.9798 versus 0.9660**), indicating that it identified a larger proportion of observations labeled as proxy high-risk. However, its lower precision (**0.7747**) indicates a greater proportion of false-positive high-risk predictions.

The Random Forest Classifier was therefore selected as the preferred classification model because it provided a stronger overall balance between precision and recall.

These results demonstrate the ability to classify the **constructed proxy label**, not actual traffic accidents. The reported performance must not be interpreted as verified accident-prediction accuracy.

3.3 Traffic Volume Regression

The regression task aimed to predict hourly traffic volume using engineered numerical features.

Two regression approaches were compared:

- Linear Regression, serving as the baseline model.
- Random Forest Regressor, providing a nonlinear ensemble approach.

The recorded MLflow evaluation results were:

Model	MAE	R ²
Linear Regression	813.7666	0.7259
Random Forest Regressor	271.3004	0.9442

The Random Forest achieved substantially lower prediction error and higher R² than Linear Regression.

This suggests that nonlinear relationships among time, weather and traffic conditions are important for predicting traffic volume in this dataset.

3.4 Model Selection and Interpretation

Based on the recorded regression performance, Random Forest was selected as the preferred traffic-volume prediction model.

The selected model was subsequently used as the basis for MLflow experiment tracking and the Task 6 deployment simulation.

However, these evaluation results were obtained using historical observations and a random train-test split. They should not be interpreted as proof of equivalent performance on future traffic conditions.

Further evaluation using chronological holdout data and condition-specific error analysis would be required before real-world deployment.

4. Unsupervised Machine Learning

4.1 Objective

Task 2 applied two unsupervised machine learning techniques to identify meaningful patterns in historical traffic conditions:

- **K-means Clustering** to group observations with similar traffic characteristics.
- **Association Rule Mining** to identify relationships among time of day, day type, weather conditions and congestion levels.

Because no real accident dataset was available, the analysis focused on traffic conditions rather than accident events.

4.2 K-means Clustering

K-means clustering was applied using three features: hour of day, weather severity and traffic volume.

Weather conditions were encoded into three severity levels, and the clustering variables were standardized before model fitting.

Candidate cluster counts from **K=2 to K=6** were evaluated using the silhouette score. The highest score was obtained at **K=6**, with a silhouette score of **0.5254**.

The final K-means model was trained on all **48,187 observations**.

4.3 Cluster Interpretation

The six clusters formed three main traffic profiles, each further differentiated by weather severity.

Traffic Profile	Clusters	Average Traffic Volume
Overnight, low traffic	2 and 4	Approximately 938–941
Evening, declining traffic	1 and 3	Approximately 2,514–2,670
Daytime, high traffic	0 and 5	Approximately 5,038–5,099

The analysis indicates that **time of day is an important factor in distinguishing traffic conditions**.

Weather severity further separates observations with otherwise similar temporal and traffic-volume characteristics.

These clusters are descriptive groupings of historical traffic conditions and do not represent accident predictions.

4.4 Association Rule Mining

The Apriori algorithm was used to identify relationships among time of day, weekday/weekend status, weather conditions and congestion categories.

The analysis generated **154 frequent itemsets** and retained **184 association rules** involving congestion outcomes.

Rules were evaluated using support, confidence and lift.

One particularly meaningful rule was:

Weekend + Night → Low Congestion

This rule achieved:

- Support: **0.0644**
- Confidence: **0.8861**
- Lift: **3.5443**

The confidence indicates that approximately **88.6%** of observations satisfying the weekend-night conditions were associated with low congestion.

The lift of approximately **3.54** indicates that this association occurred substantially more frequently than expected under independence.

Other high-lift rules associated nighttime conditions with low congestion, while weekday-afternoon conditions appeared in rules associated with higher congestion.

4.5 Findings and Practical Implications

The clustering and association-rule results consistently demonstrate the importance of time of day and day type in historical traffic patterns.

Overnight periods are generally associated with lower traffic, while daytime and weekday-afternoon conditions are more strongly associated with higher traffic volumes.

These findings provide analytical support for the travel-timing recommendation system developed later in Part 3.

However, the patterns represent historical associations rather than causal relationships. They should not be interpreted as real-time traffic forecasts or evidence of accident risk.

5. Deep Learning and Explainability

5.1 Objective

Task 3 developed a feed-forward neural network to predict hourly traffic volume and applied SHAP explainability to a comparable Random Forest Regressor.

The objective was to evaluate whether a neural network could effectively model traffic demand and to identify the features that contributed most strongly to model predictions.

5.2 Neural Network Architecture

The neural network was developed using TensorFlow and Keras with **25 engineered predictor features**.

The architecture consisted of:

- Input layer with 25 features.
- First hidden Dense layer with 64 neurons and ReLU activation.
- Second hidden Dense layer with 32 neurons and ReLU activation.
- Output layer with one linear neuron for traffic-volume regression.

The model used the Adam optimizer with a learning rate of 0.001 and Mean Squared Error (MSE) as the training loss.

Numerical features were standardized using StandardScaler.

The dataset was divided into **30,839 training observations, 7,710 validation observations** and **9,638 test observations**.

Training was configured for a maximum of 50 epochs, with early stopping to reduce unnecessary training and limit overfitting.

5.3 Model Evaluation

The neural network achieved the following performance on the held-out test dataset:

Metric	Neural Network
Mean Absolute Error (MAE)	297.8235
R-squared (R^2)	0.9395

The model explained approximately **93.95% of the variation** in the test-set traffic volume.

Training and validation losses were monitored across 50 epochs, with the best recorded validation loss occurring at epoch 50.

These results demonstrate that the neural network was capable of learning meaningful relationships between temporal, weather-related and traffic-demand variables.

5.4 SHAP Explainability

SHAP was applied to a comparable Random Forest Regressor trained on the same traffic-volume prediction problem.

The Random Forest achieved an MAE of **275.0783** and an R^2 of approximately **0.9436**, slightly outperforming the neural network.

SHAP analysis was performed on a reproducible sample of **100 test observations**. Mean absolute SHAP values were calculated to identify the features with the greatest average influence on predictions.

The five most influential features were:

Rank	Feature	Mean Absolute SHAP
1	hour_cos	1444.8046
2	hour	444.4674
3	hour_sin	188.3298
4	day_sin	147.4430
5	day_of_week	144.3160

The results indicate that **time-related features were the dominant contributors** to predicted traffic volume.

In particular, the cyclical hour representation (hour_cos and hour_sin) captured important daily traffic patterns.

Day-of-week variables also contributed to model predictions, reflecting differences in traffic behavior across the weekly cycle.

5.5 Interpretation and Limitations

The neural network and comparable Random Forest both demonstrated strong predictive performance.

However, the Random Forest achieved slightly lower prediction error and higher R^2 while requiring a less complex modeling approach.

SHAP improved interpretability by identifying which input variables contributed most strongly to the Random Forest predictions. Because SHAP was applied to a comparable tree-based model rather than directly to the neural network, its results explain the Random Forest and should not be interpreted as exact explanations of the neural network's internal behavior.

The analysis also remains limited to historical observations from a single traffic corridor. Strong test performance does not guarantee equivalent results on future or geographically different traffic conditions.

Overall, Task 3 demonstrated that deep learning can support traffic-demand prediction, while SHAP provides useful insight into the importance of temporal and weather-related features.

6. Advanced AI Technique: MLflow Experiment Tracking

6.1 Objective and Technique Selection

Task 4 implemented **MLflow experiment tracking** as the selected advanced AI technique.

MLflow was chosen because the project involved comparing multiple machine learning models and required a reproducible approach to recording model configurations, evaluation metrics and experiment histories.

Unlike techniques such as model quantization or synthetic data generation, MLflow directly supports the management and operational lifecycle of the predictive models developed in this project.

It also provides a foundation for the MLOps deployment simulation implemented in Task 6.

6.2 Implementation

MLflow was configured with a local SQLite tracking database named `mlflow_tracking.db`.

Two regression models were recorded and compared:

- **v1: Linear Regression**, serving as the baseline.
- **v2: Random Forest Regressor**, serving as the selected deployment candidate.

The tracked information included model parameters, evaluation metrics, experiment runs and execution status.

The completed experiments produced the following results:

Model Version	Algorithm	MAE	R ²
v1	Linear Regression	813.7666	0.7259
v2	Random Forest Regressor	271.3004	0.9442

The Random Forest was configured with 100 estimators and random_state=42, using an 80/20 train-test split and 25 predictor features.

The MLflow tracking database also retained an unsuccessful Random Forest run, demonstrating that experiment history can include both completed and failed attempts.

Only successfully completed runs were considered for model selection.

6.3 Value Added by MLflow

MLflow provided several benefits to the project.

Experiment reproducibility: Model configurations and evaluation results were stored in a structured tracking system rather than relying only on notebook outputs.

Model comparison: Linear Regression and Random Forest could be compared using consistent performance metrics.

Model traceability: Each experiment was associated with a unique run identifier, allowing results to be retrieved and reviewed later.

Deployment integration: The selected Random Forest experiment provided the baseline configuration and performance information used in Task 6.

Auditability: Failed and completed runs remained distinguishable in the experiment history.

These capabilities improved the organization, transparency and reproducibility of the machine learning workflow.

6.4 Limitations

MLflow experiment tracking does not automatically guarantee that a model is suitable for production deployment.

The original Task 4 Random Forest run retained parameters and metrics but did not expose an accessible deployable model artifact.

Consequently, Task 6 reconstructed a deployment candidate using the recorded model configuration and stored it in a separate deployment experiment.

The reconstructed model achieved an MAE of 270.0103 and an R^2 of 0.9448, closely matching the original Task 4 results.

Although the performance difference was small, the reconstructed model should not be described as an identical copy of the original trained model.

Furthermore, the local MLflow implementation did not provide a production model registry, access-control framework or continuously operating monitoring service.

6.5 Interpretation

MLflow successfully connected model experimentation with the later deployment simulation.

Its main contribution was not improving prediction accuracy directly, but enabling systematic experiment tracking, model comparison, traceability and reuse of recorded model specifications.

The technique therefore strengthened the overall MLOps workflow and provided an important link between model development and operational simulation.

7. Traffic Recommendation System

7.1 Objective

Task 5 developed a practical traffic recommendation system that translates historical traffic analysis into actionable travel-time guidance.

Because the dataset represents a **single westbound I-94 corridor**, the system focuses on recommending **when to travel rather than which route to take**.

The objective is to identify historically lower-traffic travel periods while considering day type, preferred travel hours and weather conditions.

7.2 Recommendation Methodology

The recommendation system uses historical hourly traffic volume to identify suitable travel windows.

The main steps are:

1. Analyze historical traffic volume by hour and day type.
2. Distinguish between weekday and weekend travel patterns.
3. Restrict recommendations to the user's preferred travel window.
4. Incorporate weather conditions when sufficient historical observations are available.
5. Require at least 30 observations for a weather-specific hourly recommendation.
6. Reject candidate periods whose average traffic volume is not below the relevant day-type average.
7. Fall back to broader day-type patterns when weather-specific evidence is insufficient.
8. Generate a plain-language recommendation.

The default practical travel period is **06:00–22:00**, although the recommendation function allows users to specify a different range.

7.3 Recommendation Results

Three representative scenarios were evaluated.

Scenario	Recommended Travel Window	Traffic Reduction
Weekday	22:00–23:00	39.8% below weekday average

Weekend	06:00–07:00	57.6% below weekend average
Weekday with Rain	22:00–23:00	43.4% below weekday average

For a weekday journey, the recommended period was **22:00–23:00**, with an average historical traffic volume of approximately **2,126 vehicles per hour**.

For a weekend journey, the recommended period was **06:00–07:00**, with approximately **1,089 vehicles per hour**.

Under rainy weekday conditions, the system recommended **22:00–23:00**, with approximately **1,999 vehicles per hour**, supported by **174 historical observations**.

7.4 Plain-Language Recommendations

The system converts analytical results into clear recommendations for non-technical users.

Weekday example:

"For a weekday journey, consider traveling between 22:00 and 23:00, when historical traffic volume is about 39.8% below the weekday average."

Weekend example:

"For a weekend journey, consider traveling between 06:00 and 07:00, when historical traffic volume is about 57.6% below the weekend average."

Rainy weekday example:

"For a weekday journey under rainy conditions, consider traveling between 22:00 and 23:00, when historical traffic volume is about 43.4% below the weekday average."

7.5 Robustness and Fallback Handling

The recommendation system includes safeguards against unreliable or misleading recommendations.

For example, the weekday fog analysis identified a supported historical period at 09:00 with an average traffic volume of approximately **4,969 vehicles per hour**.

Although the period satisfied the minimum observation requirement, its traffic volume exceeded the overall weekday average of approximately **3,533 vehicles per hour**.

The system therefore rejected the fog-specific recommendation and fell back to the broader weekday traffic pattern.

The system also validates user inputs and rejects unsupported day-type values.

7.6 Practical Value and Limitations

The recommendation engine provides practical, evidence-based travel-time guidance and demonstrates how historical analysis can support mobility decision-making.

However, the recommendations are based on historical traffic observations from a single corridor.

They are not alternative route selections, real-time forecasts, guarantees of lower traffic on an individual journey or evidence that a recommended period is safer.

The system should therefore be understood as a historical traffic decision-support tool rather than a production navigation service.

8. MLOps and Deployment Simulation

8.1 Objective

Task 6 demonstrated how the selected traffic-volume prediction model could progress from experimentation to deployment, monitoring and operational alerting.

The objective was not to establish a production traffic service, but to demonstrate a reproducible MLOps workflow using MLflow, FastAPI and simulated model-performance monitoring.

8.2 Model Versioning and Experiment Tracking

The existing Task 4 MLflow database was reused to retrieve historical regression experiments.

Two completed model versions were documented:

Model Version	Algorithm	MAE	R ²
v1	Linear Regression	813.7666	0.7259
v2	Random Forest Regressor	271.3004	0.9442

The Random Forest was selected as the deployment candidate because it achieved substantially lower prediction error than the baseline model.

MLflow recorded the experiment parameters, performance metrics, run identifiers and execution status.

8.3 Deployment Model Reconstruction

The original completed Task 4 Random Forest experiment did not retain an accessible deployment-ready model artifact.

Therefore, a deployment candidate was reconstructed using the model configuration recorded in MLflow.

The reconstructed Random Forest used:

- 100 estimators.
- `random_state=42`.
- 25 numerical predictor features.
- An 80/20 train-test split.
- No additional feature scaling.

The reconstructed model achieved:

Metric	Task 4 Baseline	Reconstructed Model
MAE	271.3004	270.0103
R^2	0.9442	0.9448

The MAE difference was approximately **0.48%**, indicating closely aligned predictive performance.

The reconstructed model was stored in a separate MLflow deployment experiment, preserving the original Task 4 experiment history.

8.4 FastAPI Deployment Simulation

A lightweight FastAPI application was developed to demonstrate how the Random Forest model could be exposed through a prediction service.

The /predict endpoint accepts the same 25 numerical features used by the model, validates the input structure and returns a predicted hourly traffic volume.

The deployment artifact was successfully reloaded from MLflow and tested using a held-out observation.

The model returned a prediction of approximately **4,659.6 vehicles per hour**, compared with an observed value of **4,924.0 vehicles per hour**.

The FastAPI endpoint also returned a valid response identifying model version v2 and the predicted traffic volume.

The API was tested locally within the notebook. This demonstrated the prediction function without requiring a public production server.

8.5 Prediction Error Monitoring

A monitoring simulation was implemented using Mean Absolute Error (MAE).

The authoritative Task 4 Random Forest MAE of **271.3004** was retained as the monitoring baseline.

The simulated alert threshold was defined as:

Alert Threshold = Baseline MAE × 1.5

This produced an alert threshold of approximately **406.95**.

Two scenarios were tested:

Monitoring Scenario	Monitored MAE	Status
Normal performance	320.0	PASS / Normal
Degraded performance	450.0	ALERT / Requires investigation

The normal scenario remained within the permitted threshold, while the degraded scenario exceeded it.

8.6 Operational Alerting

The monitoring system generated clear operational messages:

PASS / Normal – Monitored MAE: 320.0; Alert threshold: 407.0.

ALERT / Requires investigation – Monitored MAE: 450.0; Alert threshold: 407.0.

These alerts demonstrate how prediction-error monitoring can support operational decisions.

In a real deployment, an ALERT would require investigation and could trigger temporary suspension, restriction or rollback of the deployed model.

8.7 Limitations and Interpretation

The deployment system was implemented as a local simulation rather than a continuously operating production service.

The monitoring scenarios used simulated MAE values rather than errors collected from live traffic predictions.

The 1.5-times alert threshold was selected as a transparent demonstration rule and was not calibrated against real operational requirements.

Furthermore, the reconstructed deployment model demonstrated similar performance to the original Task 4 model but was not verified as a binary-identical copy.

Despite these limitations, Task 6 successfully demonstrated the key MLOps components: model versioning, experiment tracking, deployment simulation, monitoring and operational alerting.

9. Responsible and Sustainable AI

9.1 Data Bias and Coverage Limitations

The traffic intelligence solution is based on historical observations from a single westbound I-94 corridor near Minneapolis–St. Paul, covering approximately 2012 to 2018.

Consequently, the dataset does not represent an entire city, other road networks or current traffic conditions. Model performance may differ when applied to locations or time periods outside the original dataset.

9.2 Proxy Labels and Accident-Risk Limitations

No real accident dataset was used in this project.

The classification task used a documented proxy derived from traffic-volume-based congestion categories combined with severe or low-visibility weather conditions.

This proxy was designed to demonstrate the classification workflow and must not be interpreted as actual accident occurrence or verified accident probability.

Similarly, the recommendation system identifies historically lower-traffic periods rather than safer routes or lower accident risk.

9.3 Fairness and Uneven Model Errors

Overall performance metrics do not guarantee that prediction errors are evenly distributed across operating conditions.

Errors may differ between weekday and weekend periods, peak and off-peak hours, holidays and weather categories.

Rare conditions such as fog or squall contain relatively few historical observations, making their predictions potentially less reliable.

Furthermore, the random train-test split used in the regression experiments may produce optimistic estimates of performance on future observations.

Time-based validation and subgroup error analysis are therefore recommended before real-world deployment.

9.4 Governance and Human Oversight

Real-world deployment should require appropriate human review and approval.

MLflow model versioning, experiment tracking and monitoring records provide a foundation for transparency and auditability.

The Task 6 monitoring simulation demonstrated a prediction-error threshold based on 1.5 times the Task 4 Random Forest MAE.

When monitored performance exceeds the threshold, the system generates an **ALERT / Requires investigation** status.

In a production environment, such an alert should trigger human investigation and, where appropriate, temporary suspension, restriction or rollback of the deployed model.

9.5 Sustainability and Resource Efficiency

The project considered the trade-off between predictive performance and computational requirements.

The Random Forest Regressor achieved strong traffic-volume prediction performance while remaining practical to train and deploy on conventional computing hardware.

More complex neural-network models may require additional training time, memory, computational resources and electricity.

Model complexity should therefore be justified by meaningful improvements in predictive performance or operational value.

Sustainable MLOps practices include reusing validated data pipelines, avoiding unnecessary retraining and maintaining only the model artifacts and logs required for reproducibility and governance.

9.6 Responsible AI Conclusion

The project demonstrates the importance of evaluating machine learning systems beyond predictive accuracy.

Geographic and temporal coverage, proxy-label validity, uneven error distribution, human oversight, monitoring and computational sustainability must all be considered before deploying mobility AI in real-world environments.

A detailed discussion is provided in the accompanying **Responsible AI: Bias, Fairness, Governance and Sustainability Report**.

10. Integration and Conclusion

10.1 Integration of the AI Solution

Part 3 integrates supervised learning, unsupervised learning, deep learning, explainability, experiment tracking, traffic recommendations, deployment simulation and responsible AI into a coherent traffic intelligence workflow.

The solution builds on the data preparation and feature engineering developed in Part 2.

The integrated workflow consists of the following stages:

Stage 1 – Data Preparation and Feature Engineering

Historical traffic and weather observations are processed into structured numerical features suitable for machine learning. Temporal variables, cyclical encodings, weather indicators and holiday information provide a common analytical foundation.

Stage 2 – Supervised Machine Learning

Classification models identify traffic conditions using a documented accident-risk proxy, while regression models predict hourly traffic volume.

The Random Forest algorithms demonstrated strong performance in both classification and regression, supporting their selection for the corresponding prediction tasks.

Stage 3 – Unsupervised Pattern Discovery

K-means clustering identifies groups of similar traffic conditions, while association rule mining reveals relationships among time of day, weather and congestion.

These findings support the interpretation of historical traffic patterns and inform the travel-timing recommendation system.

Stage 4 – Deep Learning and Explainability

A feed-forward neural network demonstrates an alternative approach to traffic-volume prediction.

SHAP analysis of a comparable Random Forest model identifies the importance of temporal features and improves understanding of the prediction process.

Stage 5 – Experiment Tracking and Model Selection

MLflow records model configurations, performance metrics and experiment histories.

The selected Random Forest regression model achieved an MAE of **271.3004** and an R^2 of **0.9442**, providing the reference for the deployment simulation.

Stage 6 – Traffic Recommendation System

Historical traffic patterns are converted into practical travel-time recommendations based on day type, preferred travel hours and weather conditions.

The system includes observation thresholds, fallback behavior and plain-language recommendations.

Stage 7 – Deployment, Monitoring and Alerting

The Random Forest model is reconstructed from its recorded configuration, stored as a deployment-ready MLflow artifact and exposed through a FastAPI prediction endpoint.

Prediction-error monitoring and operational alerts demonstrate how model reliability could be assessed after deployment.

Stage 8 – Responsible AI and Governance

The solution incorporates explicit consideration of data coverage, proxy-label risks, uneven model errors, human oversight and sustainability.

These safeguards define the limits of the prototype and the additional requirements for responsible real-world use.

10.2 Key Project Findings

The integrated analysis produced several important findings.

First, temporal variables are central to traffic-volume prediction. Both the unsupervised analysis and SHAP explainability results indicate that hour of day and related cyclical features are strongly associated with traffic patterns.

Second, the Random Forest Regressor provided substantially better predictive performance than the Linear Regression baseline, demonstrating the value of modeling nonlinear relationships in historical traffic data.

Third, clustering and association rule mining identified meaningful traffic profiles and congestion associations that could be translated into practical travel-time guidance.

Fourth, the recommendation system demonstrated how analytical results can be communicated to non-technical users through clear, actionable recommendations.

Finally, MLflow experiment tracking, FastAPI deployment simulation, monitoring and alerting demonstrated how predictive models could be managed beyond the initial training stage.

10.3 Limitations and Future Improvements

Despite these achievements, the system has several limitations.

The dataset represents historical traffic observations from a single westbound I-94 corridor and does not support city-wide route optimization or real-time traffic prediction.

No real accident dataset was used. Accident-risk classification was demonstrated using a documented proxy based on congestion and weather conditions rather than observed accident outcomes.

The models were evaluated primarily using historical data and random train-test splits. Future work should include chronological validation, more recent observations and subgroup error analysis.

Additional improvements could include real-time traffic integration, broader geographic coverage, operationally calibrated monitoring thresholds and formal human approval processes.

10.4 Final Conclusion

The Smart City Traffic Intelligence project demonstrates how data analytics, Python engineering and machine learning can be combined to support evidence-based mobility decision-making.

Part 3 extends the earlier data preparation and analysis into a connected AI workflow covering traffic prediction, pattern discovery, explainability, recommendation generation, experiment tracking and deployment simulation.

The project demonstrates the technical feasibility of a historical traffic intelligence prototype while acknowledging the limitations of its data, proxy labels and simulated deployment environment.

Overall, the solution provides a reproducible foundation for future mobility AI development and illustrates how predictive performance, interpretability, governance

and sustainability can be considered together when designing responsible intelligent transport systems.